

Tutorial No:5

Tutorial Title: Write a C++ program to overload the interest() function to calculate Simple Interest and Compound Interest.

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Batch:C2

Program:

```
#include <iostream>
#include <cmath>
using namespace std;

// Function to calculate Simple Interest
double interest(double p, double r, double t)
{
    return (p * r * t) / 100;
}

// Overloaded function to calculate Compound Interest
double interest(double p, double r, double t, int n)
{
    return p * pow((1 + r / (100 * n)), n * t) - p;
}

int main()
{
    double principal, rate, time;
    int frequency;

    cout << "Enter Principal: ";
    cin >> principal;

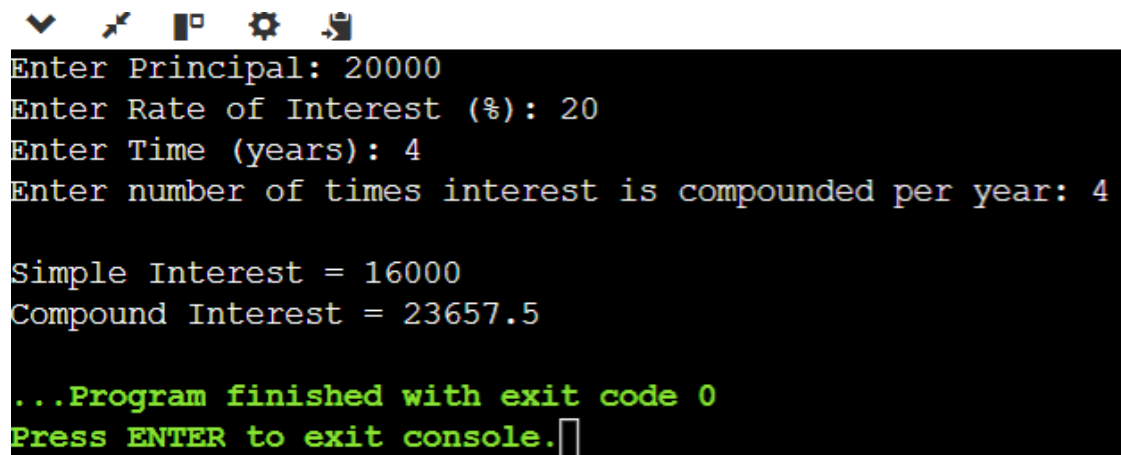
    cout << "Enter Rate of Interest (%): ";
    cin >> rate;

    cout << "Enter Time (years): ";
    cin >> time;

    cout << "Enter number of times interest is compounded per year: ";
    cin >> frequency;
```

```
cout << "\nSimple Interest = "  
    << interest(principal, rate, time);  
  
cout << "\nCompound Interest = "  
    << interest(principal, rate, time, frequency);  
  
return 0;  
}
```

Output:



The screenshot shows a terminal window with a dark background. At the top, there is a toolbar with icons for a dropdown menu, a cursor, a square, a gear, and a document. The terminal text is as follows:

```
Enter Principal: 20000  
Enter Rate of Interest (%): 20  
Enter Time (years): 4  
Enter number of times interest is compounded per year: 4  
  
Simple Interest = 16000  
Compound Interest = 23657.5  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```